

Hybrid LBP-ResNet50 Model

Image Branch + LBP
Branch + Fusion Classifier

Phase 1: Hybrid Head Training

ResNet50 Backbone Frozen

Training Configuration

Adam, Learning Rate = 1×10^{-4} , Maximum 25 Epochs

Training Control

Early Stopping: Patience = 5
ReduceLROnPlateau: Patience = 3, Factor = 0.5

Best Phase 1 Model

Selected According to Minimum Validation Loss

Phase 2: Fine-Tuning

Last 40 ResNet50 Layers Unfrozen
Batch Normalization Layers Remain Frozen

Fine-Tuning Configuration

Adam, Learning Rate = 3×10^{-5} , Maximum 15 Epochs

Training Control

Early Stopping: Patience = 7
ReduceLROnPlateau: Patience = 2, Factor = 0.5

Best Fine-Tuned Hybrid Model

Selected According to Minimum Validation Loss

Final Model Evaluation

Accuracy, Precision, Recall, F1-Score,
Confusion Matrix, and ROC-AUC